

Original Research

Precision Drug Repurposing (PDR): Patient-level modeling and prediction combining foundational knowledge graph with biobank data

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ABSTRACT

Objective: Drug repurposing accelerates therapeutic development by finding new indications for approved drugs. However, accounting for individual patient differences is challenging. This study introduces a Precision Drug Repurposing (PDR) framework at single-patient resolution, integrating individual-level data with a foundational biomedical knowledge graph to enable personalized drug discovery.

Methods: We developed a framework integrating patient-specific data from the UK Biobank (Polygenic Risk Scores, biomarker expressions, and medical history) with a comprehensive biomedical knowledge graph (61,146 entities, 1,246,726 relations). Using Alzheimer's Disease as a case study, we compared three diverse patient-specific models with a foundational model through standard link prediction metrics. We evaluated top predicted candidate drugs using patient medication history and literature review.

Results: Our framework maintained the robust prediction capabilities of the foundational model. The integration of patient data, particularly Polygenic Risk Scores (PRS), significantly influenced drug prioritization (Cohen's $d = 1.05$ for scoring differences). Ablation studies demonstrated PRS's crucial role, with effect size decreasing to 0.77 upon removal. Each patient model identified novel drug candidates that were missed by the foundational model but showed therapeutic relevance when evaluated using patient's own medication history. These candidates were further supported by aligned literature evidence with the patient-level genetic risk profiles based on PRS.

Conclusion: This exploratory study demonstrates a promising approach to precision drug repurposing by integrating patient-specific data with a foundational knowledge graph.

1. Introduction

Drug repurposing, the process of finding new applications for existing, approved drugs, offers a time and cost-efficient approach to drug development, potentially significantly reducing the traditional 10–17-year development timeline [1,2]. However, patient-centric perspective on drug repurposing remains an unexplored topic, particularly at the level of individual patients. In this study, we explore the potential of integration of patient-specific data for personalized drug repurposing (PDR), with an emphasis on understanding how individual genetic variations might impact drug discovery. While this approach is increasingly reliant on computational strategies such as machine learning and network analysis, which efficiently handle diverse data types but face challenges such as limited datasets and data interaction oversight [3], we choose to implement our method using Knowledge Graphs (KGs). KGs have emerged as an effective solution for managing and combining heterogeneous and complex semantic biomedical relationships

from various sources, enabling flexible data structuring and complex relationship analysis in pharmaceutical R&D [3,4].

In recent years, many research labs have shifted their focus on creating KGs. Ping et al. propose individualized KGs (iKGs) as a tool for cardiovascular medicine [5]. iKGs integrate biological data with patient medical histories, with the authors noting “a great need for the development of machine learning–based mapping algorithms to infer connections across data types” [5]. PrimeKG, another multimodal KG, involves relationships representing ten major biological scales [6]. DReCHGR addresses heterogeneous Electronic Health Records (EHRs) utilization for drug recommendation [7], while AIMedGraph integrates precision medicine information related to genes and genetic alterations [8]. Soman et al. [9] demonstrated that embedding EHR data into biomedical KGs outperforms raw EHR data in distinguishing Parkinson's disease patients. However, none of these approaches have explored drug

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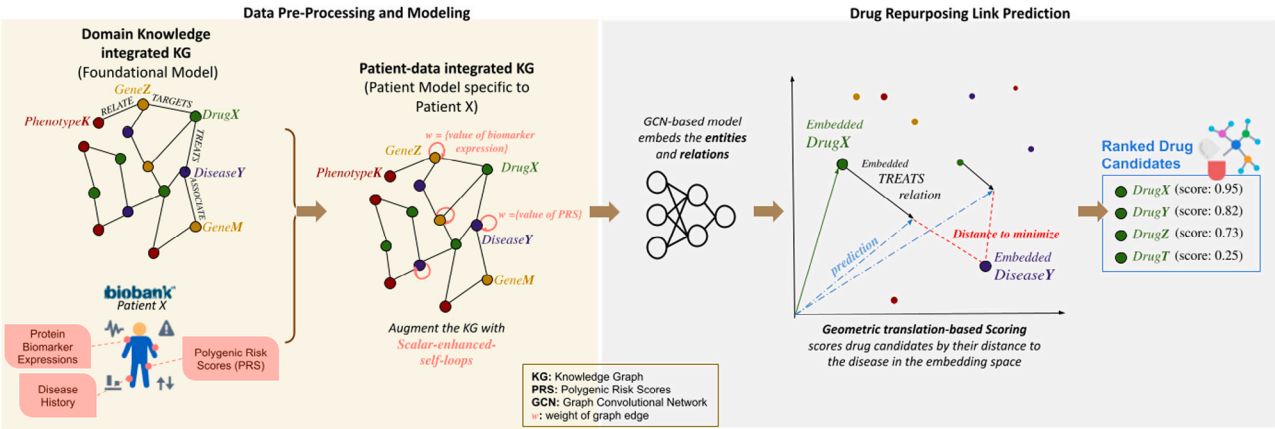


Fig. 1. The overall workflow of data integration, embedding and prediction. The data of Patient X is integrated to the KG created by domain knowledge. The embedding workflow uses a GCN-based embedding and a translation-based scoring function to get the drug–disease interaction probabilities for a given disease of interest (Disease Y). The prediction process sorts the drug–disease pairs in descending order to find the top drug candidates.

Table 1	
Statement of significance.	
Field	Description
Problem	Current drug repurposing approaches use a “one size fits all” methodology that does not account for individual patient genetic diversity and biological variations.
What is Already Known	Drug repurposing offers a faster and more cost-effective approach to drug development compared to traditional methods. Additionally, genetic factors play a crucial role in drug efficacy and safety, particularly for neurodegenerative disorders.
What this Paper Adds	This paper presents the first framework that operates at individual patient granularity, integrating patient-specific data — such as Polygenic Risk Scores (PRS), biomarker expressions, and medical history — into a foundational biomedical knowledge graph to enable personalized drug repurposing approaches.

repurposing at the granularity of a single patient. Unlike existing models, our patient-centric knowledge graph is the first to explore personalized drug repurposing by combining multi-modal data (such as protein expression, PRSs, and EHRs) at the individual patient level with a foundational knowledge graph. In this study, we experiment with a graph neural network model to incorporate patient-specific data within a unified knowledge graph, seeking to enable more tailored, individual-level drug repurposing strategies that consider genetic risk factors and patient-specific features in drug candidate prediction and prioritization (see Table 1).

2. System and methods

2.1. Foundational model

We defined two models for this study: the *foundational model* and the *patient model*. The foundational model was constructed from existing biomedical knowledge bases, while the patient model was constructed by integrating patient-specific data into the foundational model for drug–disease interaction prediction. Both models leverage the same Graph Convolutional Network (GCN) architecture.

2.1.1. Foundational knowledge graph

We utilized GP-KG, a comprehensive biomedical knowledge graph developed by Gao et al. [10], as our foundational knowledge representation framework. This knowledge graph integrates diverse biomedical

information from multiple established databases, connecting entities such as drugs, genes, diseases, and phenotypes through various semantic relationships (treat, relate, target, etc. (Fig. 2a)). In total, GP-KG contains 61,146 entities and 1,246,726 relations, organized into 7 entity types and 9 relationship types (Fig. 2). For a detailed description of the data sources and graph structure, see [10].

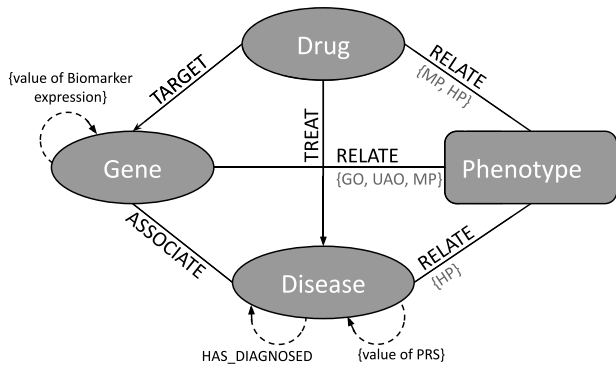
2.2. Patient specific model

2.2.1. Patient data

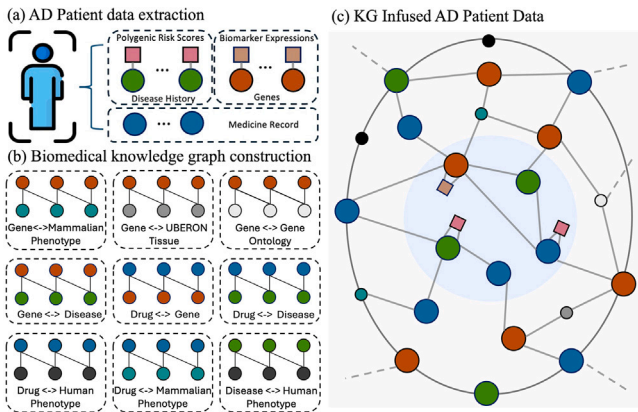
The UK Biobank, encompassing more than 500,000 participants, is the most densely phenotyped longitudinal study currently [11]. Our UK Biobank patient cohort included 4,124 participants who were aged 50–65 and diagnosed with Alzheimer’s Disease (AD). We integrated patient-level data from the UK Biobank into GP-KG.

In our case studies, we focus on AD due to its high genetic heterogeneity and the lack of precision medicine-based therapeutics [12]. This characteristic allows us to examine if our model can identify repurposed candidate drugs by incorporating patient’s biobank data, including: (i) Polygenic Risk Scores (PRS), (ii) Disease Diagnosis and History, and (iii) Protein Biomarker Expressions.

1. Polygenic Risk Scores, also known as genome-wide polygenic scores, are used to estimate an individual’s likelihood of developing specific diseases or traits based on the cumulative impact of numerous genetic variants [11]. PRS is calculated by summing the weighted risk alleles (variants) in a person’s genome [13]. The weight is determined by the effect size, usually estimated from a genome-wide association study (GWAS). The resulting score is approximately normally distributed in the general population where a higher PRS suggests an increased risk of disease. In our study, we utilized the standard PRS for Alzheimer’s Disease, along with PRS for several comorbidities, including atrial fibrillation, bipolar disorder, cardiovascular disease, coeliac disease, coronary artery disease, hypertension, multiple sclerosis, osteoporosis, Parkinson’s disease, rheumatoid arthritis, schizophrenia, and type 2 diabetes [14]. Using PRS allows for tailoring drug therapies based on patients’ genetic predispositions, improving the precision of drug repurposing.
2. The diagnosis data for each patient is compiled by combining the ‘Diagnoses - ICD10’ and ‘Diagnoses - Secondary ICD10’ fields, with any duplicates being removed.
3. We analyzed the protein biomarker expression data for 2923 entries per patient from the UK Biobank. [15] provides more details on how UK Biobank processes the data type. We examined both the up-regulated and down-regulated biomarker expressions, as well as their reference levels.



(a) **Schema** The entities and relations types of the KG are schematized. Patient's data are integrated by introducing scalar-enhanced self-loops to the corresponding entities. Drug history information of patient is withheld from the training data, to be used in *comparison of patient medication history* (2.3.2).



(b) **Flow chart** (a) extracted AD patient information from UK Biobank database, (b) extracted raw interactions from biomedical databases and integrated these interactions into a KG. (c) infused AD patient data into KG.

Fig. 2. Visualizations of patient data infused KG.

This retrospective study is exempt from informed consent as it involves a secondary analysis of existing, de-identified data without human subject interaction or intervention.

2.2.2. Embedding algorithm

In this section, we show the details of the embedding process including message passing and aggregation, self-loops and scalar modulation of relation embeddings. We approach drug repurposing as a link prediction task in knowledge graphs, where the goal is to identify and rank potential therapeutic relationships between drugs and diseases (See A.1 for formal problem definition). This ranking is personalized through the integration of patient-specific data, as detailed below.

We integrate patient-specific data into the knowledge graph through two mechanisms: self-loops and scalar modulation of relation embeddings (Fig. 2). First, we add self-loops to entity nodes that correspond to patient features, allowing these characteristics to be incorporated during neighborhood feature aggregation in graph convolution. Second, we incorporate quantitative patient data (such as biomarker values and polygenic risk scores) by using them to modulate relation embeddings

during the embedding computation:

$$\text{Scalars}[i] = \begin{cases} p(v), & \text{if edge } (h, r) \text{ is a self-loop at node } v \\ 1, & \text{otherwise} \end{cases} \quad (1)$$

where $p(v)$ represents the patient-specific scalar value for node v . These scalar values dynamically adjust the relation embeddings (h_r) during the model's forward pass:

$$h_r^{k+1} \leftarrow h_r^{k+1} \odot \text{Scalars}^\top \quad (2)$$

For example, in the case of Alzheimer's Disease (AD), we add a self-loop to the AD node (entity) weighted by the patient's polygenic risk score (PRS). This ensures the patient's genetic predisposition to AD influences the drug ranking process when evaluating potential treatments like Memantine.

Building on our patient-integrated knowledge graph, we predict personalized drug-disease associations using a combination of Graph Convolutional Networks (GCNs) and translation-based scoring (as depicted in Fig. 1). Our method employs Compositional Graph Convolutional Networks (CompGCN) [16] to learn embeddings of entities (drugs, diseases) and their relationships, while efficiently handling the patient-specific features through reduced parameterization. For prediction, we utilize InteractE [17], which processes these embeddings through geometric transformations and convolution operations to generate interaction scores. These scores indicate the likelihood of therapeutic relationships between drug-disease pairs, now informed by patient-specific data. For detailed mathematical formulations, see Appendix A.2.

2.3. Comparison

2.3.1. Experimental setup

We compared the patient model with the foundational model in link prediction. The goal was to demonstrate that incorporating patient-specific data did not compromise the general knowledge reasoning capabilities of the model.

Training and evaluation setup. We conducted a robustness analysis to ensure that personalization maintains rather than degrades the foundational model's performance. Data was split into training (70%), validation (15%), and test (15%) sets using stratified sampling, maintaining entity overlap with the training set. Performance was assessed via 5-fold cross-validation. Negative samples were generated by replacing diseases in true $(V_{\text{drug}}, R_{\text{TREAT}}, V_{\text{disease}})$ triples with random diseases not linked to the drug. We used a margin-based ranking loss for optimization, where the model was trained to score true triples higher than negative triples. The Adam optimizer was employed.

Hyperparameter selection. Hyperparameters were fine-tuned via grid search using validation set's MRR, guided by prior studies [10] to ensure efficient model performance while balancing computational cost. More details on the hyperparameters are presented in Appendix A.4.

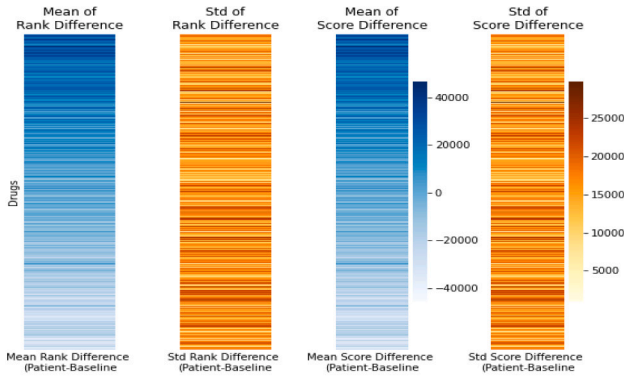
Model performance metrics. We evaluated the model using ranking metrics (Mean Rank (MR), Mean Reciprocal Rank (MRR), Hits@k for $k = 1, 3, 5, 10$) and Area Under the Curve (AUC) (Table 3) to assess how well *true* drug-disease triples were ranked against false ones. This ranking-based approach aligns with drug repurposing needs, where identifying top predictions is crucial.

2.3.2. Case study

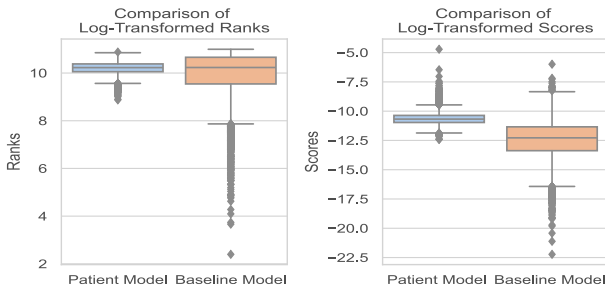
To evaluate our personalized drug repurposing approach, we conducted case studies using data from three representative Alzheimer's Disease patients (*Robin, Zoro, Usopp*) selected from the UK Biobank cohort through clustering analysis. The detailed patient selection methodology is described in Appendix A.3. Since there was no gold-standard of personalized AD drugs tailored for specific patients, we employed two strategies to evaluate top-ranked candidates (Fig. 4):

Table 2
Evidence and mechanism of the Top 3 drug candidates for Alzheimer's disease.

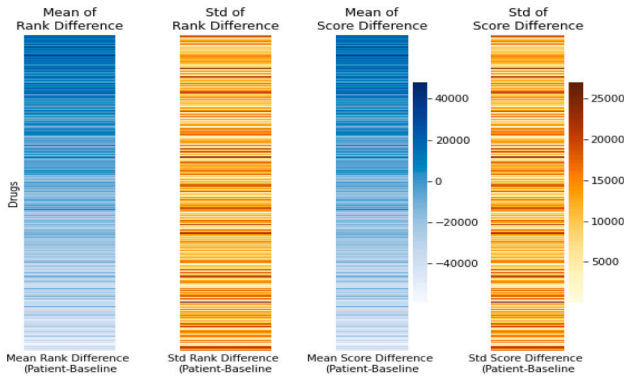
Drug Candidates	Drug Mechanism	Patient-specific Interpretation	Evidence for AD
Patient Model (Robin)			
9-Hydroxy-8-Methoxy-6-Nitro-Phenanthrol[3,4-D][1,3]Dioxole-5-Carboxylic Acid	This chemical is a derivative of aristolochic acids, with one H replaced by OH. Plants containing aristolochic acids are used in some herbal products intended to treat symptoms and diseases, such as arthritis, gout, and inflammation, kidney and urinary problems, snakebites, and other ailments.	–	There is no direct research about the influence of this chemical on Alzheimer's disease (AD). But existing research showed that aristolochic acids may cause AD [18].
S-NONYL-CYSTEINE	This chemical is a derivative of cysteine, with the H in the functional group -SH supplemented by a long carbon hydrogen chain. Cysteine is a non-essential amino acid and could be used as nutritional supplements or drugs (for malnutrition, hair care, digestion problem, chronic inflammation, type2 diabetes etc.)	–	No evidence found for AD.
Lornoxicam	Lornoxicam is a nonsteroidal anti-inflammatory drug (NSAID) indicated in the treatment of mild to moderate pain, rheumatoid arthritis and osteoarthritis. Lornoxicam is a potent inhibitor of the cyclooxygenase (COX) enzymes, which are responsible for catalyzing the formation of prostaglandins (a messenger molecule in the process of inflammation) and thromboxane. Inhibition of prostaglandin and thromboxane synthesis has anti-inflammation, antipyretic, antithrombotic and analgesic effects.	This precision prescription is consistent with the patient's previous prescription of drug DB00316 for pain management and antipyretic treatment.	Clinical trial NCT01117948
Patient Model (Zoro)			
6-bromoindirubin-3'-oxime	The synonym of the chemical is (2Z,3E)-6-bromoindirubin-3'-oxime. It is a GSK-3 β inhibitor and inhibit ovarian cancer development in ovarian cell lines in vitro.	–	No evidence found for AD.
Plozalizumab	Plozalizumab is a novel humanized monoclonal antibody and has been investigated for atherosclerosis treatment. The drug specifically targets CCR2 chemokine receptors on the surface of certain white blood cells including macrophages and monocytes. CCR2 plays an important role in the trafficking of monocytes and macrophages to sites of inflammation. The recruitment of macrophages to the arterial wall is believed to be a critical step in the development of atherosclerosis.	–	No evidence found for AD.
Ambrisentan	Ambrisentan is a selective type A endothelin receptor antagonist used to treat primary pulmonary arterial hypertension and pulmonary arterial hypertension associated with connective tissue diseases. The precision prescription is consistent with the fact that this patient have a high PRS for hypertension.	Cerebral hypoperfusion, an early stage in dementia, may worsen with ETA receptor activation [19]. Ambrisentan, an ETA receptor antagonist, may improve the cerebral hypoperfusion-related AD.	
Patient Model (Usopp)			
(2e)-N-Allyl-4-[3-(4-Bromophenyl)-5-Fluoro-1-Methyl-1h-Indazol-6-Yl]Oxy-N-Methyl-2-Buten-1-Amine	This chemical belongs to the class of organic compounds known as phenylpyrazoles. Phenylpyrazoles is a newer class of chemical pesticides.	–	No evidence found for AD.
Telcagepant	Telcagepant treats migraine. It is an antagonist of calcitonin gene-related peptide (CGRP) receptor. CGRP and its receptors are found in areas of the central and peripheral nervous system. During migraine attacks, CGRP activates these receptors and facilitates the transmission of pain impulses.	–	CGRP antagonists might be a therapeutic target to attenuate the pathological cascade and delay cognitive decline of AD in humans [20].
Lidocaine	Lidocaine could act as a local anesthetic that numbs the sensations of tissues such as in surgical procedures. It elicits its numbing activity by blocking sodium channels so that the neurons of local tissues that have the medication applied on are transiently incapable of signaling the brain regarding sensations.	This precision prescription makes sense since this patient has high PRS for rheumatoid arthritis and the drug can release the pain of arthritis.	Lidocaine has neuroprotective effect and improves memory in streptozotocin induced AD model [21].
Foundational Model			
Tetrahydrodeoxyuridine	Tetrahydrodeoxyuridine is a synthetic pyrimidine nucleoside analog and it has been used in trials studying the treatment of neoplasms, lung neoplasms, breast neoplasms, sickle cell disease, and head and neck cancer, among others.	–	No evidence found for AD.
rNAPc2	Recombinant Nematode Anticoagulant Protein c2 (rNAPc2) is a potent inhibitor of the factor VIIa/tissue factor (fVIIa/TF) complex that requires the prerequisite binding to zymogen or activated factor X. The drug has been investigated for use/treatment in thrombosis (deep vein), angina, coronary artery disease, and colorectal cancer.	–	No evidence found for AD.
(S)-iclaprim	This chemical belongs to the class of organic compounds known as 1-benzopyran. Some benzopyrans have shown anticancerous activity in vitro.	–	(S)-iclaprim belongs to benzopyran and benzopyran based compounds could be useful to treat AD [22].



(a) **Case Study for AD.** For Rank Difference the T-statistic is 21.8245, p value is 0, Cohen's d is 0.2505. For Score Difference T-statistic is 91.4281, p value is 0, Cohen's d is 1.0494.



(b) **The Boxplot Analysis** shows that the patient model assigns higher and more variable ranks and scores to drugs than the foundational model. In contrast, the foundational model exhibits more consistency but lower overall rankings and scores.



(c) **Ablation Study for Polygenic Risk Scores (PRS).** To assess the effect of integrating PRS values for patient in our patient model, we experimented with removing PRS values from our data model. For Rank Difference the T-statistic is -35.0468, p value is 0, Cohen's d is -0.4023. For Score Difference T-statistic is 67.2998, p value is 0, Cohen's d is 0.7724.

Fig. 3. Ranking and scoring differences between foundational model and patient model for each drug in case study for AD.

Comparison of patient medication history. We compare the ranking performance of our patient-specific models against the foundational model using patients' known medication histories as test sets. This comparison evaluates whether incorporating patient-specific data improves

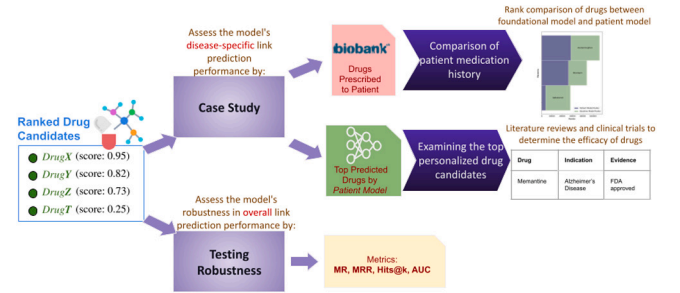


Fig. 4. The drug candidate evaluation workflow. We present our methods to evaluate predicted drug repurposing candidates and explain their objectives.

the ranking of medications already prescribed to the patient which is serving as a quantitative measure of personalization effectiveness. While medication history is not integrated into our knowledge graph (which uses only PRS, diagnosis, and biomarker data), it provides an independent validation set. This analysis also addresses the challenge of extremification [23,24], as a model that better prioritizes a patient's actual prescribed medications demonstrates balance between personalization and established treatment practices.

Examining the top personalized drug candidates. Filter bubbles represent a key challenge in precision medicine where recommendations become isolated within a narrow therapeutic space defined by the patient's data [23,24]. To detect and mitigate such bubbles, we conducted comprehensive literature searches with our domain expert for each top-ranked drug candidate, examining:

- Drug's mechanism of action in AD pathology
- Patient-specific interpretation linking drug effects to individual biomarker, PRS, and clinical data
- Supporting evidence from clinical trials and literature

This literature-based comparison reveals whether the predictions of the patient models' are confined to a limited therapeutic space or successfully identify diverse treatment opportunities supported by clinical evidence.

3. Results

3.1. Model robustness

Table 3 compares the performance of the three patient-integrated models against the foundational model using standard link prediction metrics: Hits@k, Mean Rank (MR), Mean Reciprocal Rank (MRR), and AUC. Our analysis demonstrates that incorporating patient-specific data maintains the robust knowledge reasoning capabilities of the foundational model, which has previously shown superior performance compared to traditional methods like DeepWalk and matrix factorization [10]. The comparable performance metrics between patient models and foundational model, evidenced by similar mean values and low standard deviations across patients, indicates that **personalizing the model preserves its general link prediction capabilities**. This is particularly significant because we evaluated performance across all disease domains, not just Alzheimer's Disease, confirming that our patient integration approach does not compromise the foundational model's broader knowledge reasoning power. The analysis suggests that incorporating patient-specific features provides additional insights while preserving the prediction capabilities of the base model.

Table 3

Evaluation metrics for foundational and patient-specific models. Robin, Zoro, and Usopp are the place-holder names of the patients we chose for the case studies.

	MR	MRR	Hits@1	Hits@3	Hits@5	Hits@10	AUC
Foundational Model							
	1390.6	0.19231	0.13126	0.20721	0.24959	0.31331	0.866
Patient Model							
Robin	1307.2	0.18971	0.129551	0.20414	0.24546	0.30892	0.865
Zoro	1332.3	0.19039	0.13008	0.20509	0.24692	0.30967	0.876
Usopp	1384.2	0.19636	0.13511	0.2116	0.2534	0.3181	0.826
mean	1341.2	0.19215	0.13158	0.20694	0.24859	0.31223	0.856
std	39.26962	0.003659	0.00307	0.00406	0.00423	0.00510	0.027

3.2. Effect of integrating patient data

We analyzed how patient data integration affects drug prioritization by comparing rankings and probability scores between patient-specific and foundational models. For each drug candidate, we computed the differences in rankings and scores between patient models and the foundational model, visualized in Fig. 3. The comparison revealed two key findings (Fig. 3a):

(i) **Drug Rankings:** While statistically significant (T-statistic = 21.82), the difference in drug rankings showed a small effect size (Cohen's $d = 0.25$). This suggests that patient models maintain the general ranking structure of the foundational model while making subtle but meaningful adjustments for personalization.

(ii) **Probability Scores:** The large effect size in score differences (Cohen's $d = 1.05$) indicates that patient models substantially recalibrate the confidence in drug candidates based on patient-specific features. This demonstrates that our patient integration approach meaningfully influences the model's assessment of drug-disease relationships when considering individual patient characteristics.

3.3. Ablation study for Polygenic Risk Scores

To understand the impact of Polygenic Risk Scores (PRS) on our personalized drug repurposing approach, we conducted an ablation study comparing model performance with and without PRS data (Fig. 3b). The results revealed two critical findings:

(i) **Drug Rankings:** Removing PRS altered the relationship between patient and foundational models by shifting the T-statistic from 21.82 to -35.05 . This reversal from positive to negative indicates that PRS substantially influences how patient models prioritize drugs. Without PRS, patient models not only ranked drugs differently but showed a stronger divergence from the foundational model.

(ii) **Probability Scores:** The Cohen's d for score differences decreased from 1.05 to 0.77, suggesting that PRS plays a key role in how patient models assess drug candidates. While the models still maintain distinct scoring patterns without PRS, the reduced effect size indicates that genetic risk information significantly contributes to the model's personalization capabilities.

Our case studies (Table 2) provide clear examples of this influence: Ambrisentan, typically used for pulmonary arterial hypertension, was prioritized for an Alzheimer's patient with a high PRS for hypertension (0.825732). Similarly, Lidocaine was highly ranked for a patient with elevated genetic risk for arthritis (PRS: 1.70882), suggesting its potential value not just as a local anesthetic but also for addressing inflammation risks indicated by the patient's genetic profile. These examples illustrate how PRS integration helps identify therapeutic candidates that address both the primary condition and genetically-indicated comorbidity risks.

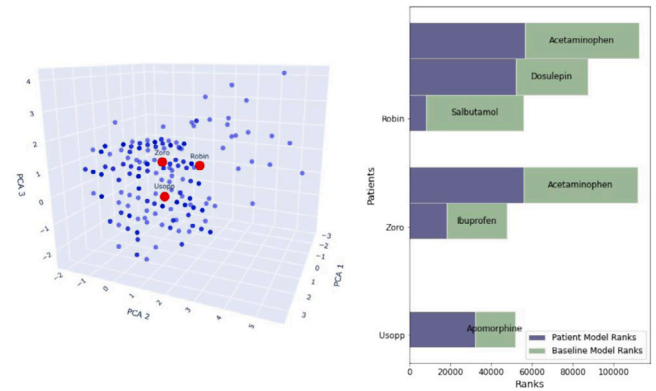


Fig. 5. Results of Comparison of Patient Medication History. A comparison of the foundational model's ranking of drugs already prescribed to patients and that of the patient model's is presented. The patient model prioritizes these drugs more than the foundational model, as can be observed from lower ranks.

3.4. Case studies

3.4.1. Comparison of patient medication history

From the patients' electronic health record from the UK Biobank, we know that the drugs each patient was prescribed with. We kept this information separate from our training and observed that (Fig. 5) the patient model prioritizes the drugs already known to be prescribed to the patients by ranking them lower, than foundational model does. This demonstrates that the patient model's outputs significantly correlate with actual clinical outcomes. We observed that the patient model generally assigned different rankings to previously prescribed drugs compared to the foundational model, which warrants further investigation into its alignment with clinical prescribing patterns.

3.4.2. Examining the top personalized drug candidates

In this exploratory PDR study, we examined potential drug repurposing candidates for each patient and investigated their biological relevance. Table 2 presents the top three novel drug repurposing candidates for Alzheimer's Disease (AD) as identified by the patient models and the foundational model, including their mechanisms, patient-specific interpretations, and supporting evidence.

For each patient, we identified drug candidates with two key features: (i) **strong evidence supporting repurposing potential**, and (ii) **mechanisms aligning with the patient's genetic risk profile based on their Polygenic Risk Scores (PRS)**.

Three examples highlight our personalized approach: **Lornoxicam for Patient Robin** aligns with their previous prescription of acetaminophen, as both drugs target the COX pathway involved in AD development. **Ambrisentan for Patient Zoro**, is typically used for pulmonary arterial hypertension. Patient Zoro shows high genetic susceptibility to hypertension (top 10 percentile PRS), addressing potential brain hypoperfusion in AD. **Lidocaine for Patient Usopp**, who has high genetic risk for rheumatoid arthritis, given the drug's mechanism of blocking sodium channels that contribute to joint pain. Detailed evidence supporting these matches is provided in Table 2.

Notably, while Lidocaine ranked within the foundational model's top 3%, Ambrisentan and Lornoxicam ranked only at the 81st and 66th percentiles, respectively. The personalized models identified therapeutic candidates that differed from those prioritized by the foundational model, suggesting complementary insights from patient-specific data.

4. Discussion

In this exploratory study, we introduce Precision Drug Repurposing (PDR) as a potential pathway to move beyond the traditional

Table 4
Glossary of abbreviations.

Abbreviations	Meanings
PDR	Precision Drug Repurposing
EHR	Electronic Health Records
AD	Alzheimer's Disease
PRS	Polygenic Risk Score
KG	Knowledge Graph
GP-KG	Name of our knowledge graph
GCN	Graph Convolutional Network
RGCN	Relational Graph Convolutional Network
MR	Mean Rank
MRR	Mean Reciprocal Rank

'one size fits all' approach to drug repurposing. Our investigation demonstrates how integrating patient-specific biobank data with foundational biomedical knowledge through graph machine learning could have potential for personalized drug repurposing while maintaining robustness of the foundational model. While we explored this approach using a knowledge graph as our foundational model and UK biobank data for patient-specific data, our framework has broader applicability to other foundational models. Through various comparison approaches, we observed that meaningful personalization could be achieved while preserving foundational model performance. Particularly intriguing was the role of Polygenic Risk Scores (PRS) in capturing patient-specific genetic predispositions that influence drug recommendations.

Our Alzheimer's disease case studies serve as preliminary examples of the framework's potential for personalizing treatment strategies based on individual genetic profiles and risk scores. While these initial cases suggest promising directions for PDR's utility, we recognize the need for broader comparison across diverse patient cohorts to better understand how genetic factors might influence precision medicine approaches.

As we explore this emerging field, several challenges have become apparent. Privacy considerations in handling patient-specific data call for robust privacy-preserving techniques. We also observed representation issues in clinical trials and potential biases in medical records that pose significant challenges [23,24]. These initial findings suggest the need for more sophisticated methods to integrate patient data into domain knowledge. Training patient-specific models is computationally intensive due to the need to integrate multiple layers of individual-level data into personalized knowledge graphs. Given these computational demands, scaling to the full UK Biobank cohort of over 500,000 participants would be impractical at this exploratory stage. This motivated our strategic selection of three representative patients through clustering techniques.

Looking ahead, our exploratory work opens various avenues for future investigation, including expanded validation across larger and more diverse patient populations. The development of more efficient data integration methods remains an important area for exploration, as does the validation of PDR-identified drug candidates through clinical trials. The insights generated by our initial precision repurposing model might inform future trial design, particularly in defining participant selection criteria. While preliminary, our work suggests promising directions for PDR while highlighting the importance of balancing individual patient characteristics with broader therapeutic knowledge.

5. Glossary

See Table 4.

CRedit authorship contribution statement

Çerağ Oğuztüzün: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal

analysis, Data curation, Conceptualization. **Zhenxiang Gao:** Writing – review & editing, Methodology, Investigation. **Hui Li:** Writing – review & editing, Methodology, Investigation. **Rong Xu:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that there are no conflicts of interest regarding the publication of this paper.

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Appendix A

A.1. Knowledge graph definition

A KG is a directed, labeled graph where nodes are connected by multiple, distinct edges [25]. Formally, a KG is represented as $KG = (V, L, F)$, with each vertex representing an entity. Each label represents a type of relation, while L represents a set of vertices. $F \subseteq V \times L \times V$ is an edge set that shows relationships as triples $\langle h, r, t \rangle$, with h as the head node, r as the relation label, and t as the tail node. For example, the triple: *Memantine* $\xrightarrow{\text{TREATS}}$ *Alzheimer's Disease* represents a relationship where $h = \text{Memantine}$, $r = \text{TREATS}$, and $t = \text{Alzheimer's Disease}$.

A.1.1. Link prediction for drug repurposing

Drug repurposing can be formulated as a link prediction task in our patient-integrated knowledge graph. Given a query $\langle h, r, ? \rangle$ (e.g., *Memantine* $\xrightarrow{\text{TREATS}}$?), our goal is to rank all potential disease targets, with the most promising therapeutic relationships ranked highest. The model evaluates the likelihood of unobserved triples $\langle h, r, t \rangle$ in the knowledge graph $KG = (V, L, F)$, where the ranking is now influenced by patient-specific features incorporated in our previous steps.

A.2. GCN-based prediction

We used a workflow that merges a GCN-based embedding and a translation-based scoring function (Fig. 1). To embed both the entities and relations of our KG, we stack CompGCN layers [16], which is an extended version of Relational Graph Convolutional Networks (RGCNs). k-stacked CompGCN performs a composition operation ϕ over each edge in the neighborhood $N(v)$ of a central node v that we want to embed as:

$$h_v^{k+1} = f \left(\sum_{(u,r) \in N(v)} W_{dir(r)}^k \phi(h_u^k, h_r^k) \right) \quad (3)$$

where $W_{dir(r)} \in \mathbb{R}^{d_1 \times d_0}$ is the relation-type specific matrix. The composed embeddings are convolved with specific filters for ingoing and outgoing edges: $W_{dir(r)}$ which can be W_O if $r \in R$, W_I if $r \in R_{inv}$, and W_S if $r = \text{self-loop}$. Then the messages from all the neighbors are aggregated f for updated embedding of the central node. Similar to RGCN, a separate weight matrix is used for each relation type, relation r is embedded after k layers as:

$$h_r^{k+1} = W_{rel}^k h_r^k \quad (4)$$

By this, they jointly embed both nodes and relations in the graph whereas RGCNs were only embedding nodes due to over-parameterization.

After completing the embedding process, we utilize InteractE [17], a translation-based model, to score and rank each drug-disease pair. The goal of translation-based scoring functions is to have the embedding of

the head entity and relation be as close as possible to the tail entity's embedding. Our input comprises the embedded drug entity h and the embedded *TREAT* relation r , with the objective being to predict a disease entity that yields the highest score according to InteractE's function:

$$\max \psi(h, r) = g(\text{concat}(f(\varphi(P_k) \otimes w)))W \quad (5)$$

where W is a learnable weight matrix. InteractE operates by generating t -random permutations of both entity Z_v and relation Z_r embeddings as $P_k = [(Z_v^1 Z_r^1); \dots; (Z_v^k Z_r^k)]$. Subsequently, it employs a chequered reshaping function $\varphi(P_k)$ to optimally capture interactions between these embeddings. The reshaped matrices of each permutation $\varphi(Z_v^i Z_r^i)$ are then layered $f(\cdot)$ into a 3D tensor, followed by depth-wise circular convolution $\otimes$. The outputs from each convolution are flattened and concatenated into a single vector. For each disease entity u , we multiply its embedding Z_u with this vector to calculate the score function $g(\cdot)$. This score is then processed through a sigmoid function to determine the probability of the drug–disease interaction triplet.

A.3. Selection of patients

We selected a group of patients for a detailed comparison with foundational model, aiming to demonstrate the effectiveness of our approach through additional case studies. This section outlines our selection process, focusing on choosing patients who represent the Alzheimer's disease population within the UK Biobank. Since this is a patient-specific model we also investigated the differences in which drugs are recommended across patients.

We were using the patient's (i) protein biomarkers, (ii) medical history with drug prescriptions and past diagnosis, (iii) PRS of comorbidities of the patient, (iv) lifestyle and demographic information of the patient (although not integrated into our data model).

For each of these categories C we applied k-means clustering within the patient cohort P , where $P = \{p_1, p_2, \dots, p_n\}$ and each patient p_i is a vector features, we partition P into clusters $S = \{S_1, S_2, \dots, S_k\}$ by minimizing the within-cluster sum of squares (WCSS) as:

$$\text{Minimize } WCSS = \sum_{i=1}^k \sum_{p \in S_i} \|p - \mu_i\|^2 \quad (6)$$

where μ_i is the centroid in S_i . We determine the optimal number of clusters k using the elbow method, by plotting the WCSS against different k s and selecting the k at which the rate of decrease changes sharply (see the elbow plots in our Supplementary Files). Consequently, each patient was assigned to a cluster within their respective group labeled as l_i , for each category. Subsequently, we merged these initial clusters and re-clustered the patients based on their previous cluster assignments, resulting in a natural final grouping of 3 distinct clusters $\{F_1, F_2, F_3\}$. To ensure a representative sample of the UK Biobank's Alzheimer's disease population, we selected one patient from each cluster, prioritizing those who were most differentiated from each other based on Euclidean distance. For each final cluster F_i we select one patient p_j^* where $d(p, \mu_m)$ is the Euclidean distance between patient and centroid of cluster F_m :

$$p_j^* = \arg \max_{p \in F_j} \left(\sum_{\substack{m=1 \\ m \neq j}}^6 d(p, \mu_m) \right) \quad (7)$$

Specifically, we calculate the distance of each point in a cluster to the centroids of all other clusters, summing these distances to determine how different a given patient is from patients in other clusters. The patient with the highest total distance to the centroids of the other clusters is considered the most different within its cluster. This method ensures that the selected patient p_j^* from each cluster is the one that is most distinct compared to patients in the other clusters, based on their initial cluster assignments. This selection process yielded 3 patients $\{p_1^*, p_2^*, p_3^*\}$ that exemplify the diversity within the studied population.

Therefore, we conducted case studies on these 3 patients (Fig. 5) and aggregated their findings for evaluation in our case study section. In order to perform *Comparison of patient medication history*, we decided not to choose patients who have no past drug prescription information. To enhance data privacy, we replaced patient IDs with fictional names (*Robin, Zoro and Usoop*).

A.4. Hyperparameter selection

- **Learning rate η :** 0.001 (from 0.0001, 0.001, 0.01) for stable optimization and fast convergence
- **Embedding dimensions κ :** 200 (from 100, 200, 400); higher values improved performance slightly but at excessive computational cost
- **GCN layers l :** 1 (from 1, 2, 3); additional layers increased cost without performance gains
- **Batch size β :** 128 (from 64, 128, 256) for balanced efficiency and performance
- **Dropout rates:** 0.1 (embedding) and 0.3 (prediction) to prevent overfitting

Appendix B. Supplementary data

Supplementary information is available in project repository under *Supplementary Material* folder https://github.com/ceragoguztuzun/PrecisionDrugRepurposing/tree/main/Supplementary_Material.

Data availability

Data used in this study are currently available on UK Biobank (<https://www.ukbiobank.ac.uk/>).

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